

Do Safe Choices Survive Encoding?

Auditing and Repairing Safe-Action Sufficiency in Pretrained Visual Representations

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Abstract

Pretrained visual representations increasingly serve as the state interface for planning, control, and safety filters. Their native objectives preserve pixels, semantic structure, or predictable features, but none requires histories mapped to one code to retain a common safe action. We call this requirement *safe-action sufficiency*. We characterize exact static and common-action obstructions, prove that deterministic post-processing cannot repair exact aliases, and give a smooth construction in which average reconstruction and prediction losses can be driven arbitrarily close to zero while worst-case safety ambiguity remains fixed. We then define a traceable finite-radius common-action audit and propose *Common-Action Geometry Repair*: update pre-bottleneck encoder modules using an action-profile loss, a set-wise neighborhood objective, and a family-specific retention loss. Unlike a safety probe, the intervention will be evaluated after removing its training head and fitting a fresh downstream readout. The prospective study deliberately triangulates reconstructive, predictive, and semantic pretraining through Cosmos Tokenizer, V-JEPA 2, and DINOv3. The protocol is designed to match the causal information span and an explicitly quantized capacity proxy within each comparison, and will use simulator-branched action profiles in manipulation and embodied navigation; driving is an optional external-validity study rather than the paper’s identity. Exact cart and pendulum aliases presently validate only the obstruction and observation-floor controls. No pretrained-encoder repair or closed-loop foundation-model result is claimed before the prospectively specified study is executed.

1 Introduction

Pretrained visual representations make physical decision-making from pixels tractable, but their compression and invariances are not neutral. Cosmos Tokenizer is optimized as a reconstructive image/video codec [19, 20]; V-JEPA 2 learns predictable spatiotemporal features and its action-conditioned variant supports planning [4, 15]; DINOv3 emphasizes semantic and dense spatial features [25]. These objectives preserve different notions of utility, yet none directly requires the representation to preserve *which action is safe*. A factor can be cheap for pixels, semantics, or average prediction and still determine whether a robot should move left, right, stop, or release an object.

This matters upstream of the controller. Recent work designs barrier functions, reachability values, or robust controllers directly in learned latent spaces [3, 13, 18, 23]. Lutkus et al. [13] define a latent safe set containing only codes with no unsafe preimage. Once an upstream code has identified two physically different histories, however, no deterministic head can recover the lost distinction. The resulting question is not only whether a code separates safe from unsafe states, but how much useful safe region and control authority it has made unavailable.

A static safe/unsafe collision is only the first obstruction. Encoding the known scalar safety margin itself separates the labels, yet states with the same margin can have different dynamics and disjoint safe actions. Under partial observability, a single image may not even identify velocity or an occluded hazard. We first ask whether one representation-measurable decision can satisfy a declared support-wise condition. Recursive representation-policy viability, Markov latent dynamics, and persistent closed-loop safety require additional theory and are not established in the main manuscript.

The scientific unit of study is therefore not autonomous driving or any one tokenizer. It is the mapping from a causal visual history to the feasible-action correspondence. We compare three families as a delib-

erately controlled objective triangulation—reconstructive compression, temporal prediction, and semantic invariance—rather than attributing causal effects to checkpoints that also differ in architecture, scale, and pretraining data. Manipulation and embodied navigation provide the primary, visually distinct control domains; driving remains a possible third stress test.

The present draft contributes: (i) a specification-relative definition of *safe-action sufficiency* and a traceable neighborhood defect that distinguishes physical infeasibility from representation-induced conflict; (ii) supporting exact results for static separation, post-processing, stochastic codes, deterministic and randomized actions, and a smooth average-loss separation; (iii) *Common-Action Geometry Repair*, a pre-bottleneck encoder intervention that combines action-profile supervision with a set-wise common-action objective and family-specific utility retention; and (iv) a prospectively specified cross-family evaluation protocol that matches causal information and an operational capacity proxy, controls readout and downstream selection, and compares Cosmos Tokenizer, V-JEPA 2, and DINOv3. Exact pixel cart and pendulum controls currently validate only the finite obstruction and observation floor. The foundation-encoder audit, repair, and downstream claims remain prospective, and every status is governed by the claim registry.

2 Related work and novelty boundary

Pretrained visual representations. Cosmos provides causal image/video tokenizers trained for reconstructive and perceptual fidelity at several spatial, temporal, continuous, and discrete compression settings [19, 20]. V-JEPA 2 and V-JEPA 2.1 learn dense predictive video features; V-JEPA 2-AC adds an action-conditioned predictor for planning while retaining the pretrained visual encoder [4, 15]. DINOv3 instead learns semantic and dense spatial features from images [25]. We use these families to span reconstructive, predictive, and semantic objectives. Their native metrics are not commensurate, and checkpoint comparisons do not identify a causal effect of objective without a matched-backbone/data ablation.

Task-aware physical tokenization. Recent work already argues that reconstruction-only driving tokens can omit planning-relevant information and augments tokenizer learning with representation and geometry supervision [31]. We therefore do not claim the first task-aware, planning-aware, or safety-augmented tokenizer. Our narrower target is a specification-relative common-safe-action obstruction, a traceable set-wise audit before policy learning, and internal encoder repair whose effect survives removal of the training head in more than one physical domain.

Latent control certificates. Latent CBF and world-model methods learn certificates or safety policies from observations [3, 16, 17]. Lutkus et al. give approximate-conjugacy conditions for transferring latent Lyapunov and barrier guarantees and explicitly identify shared safe/unsafe codes as a source of conservative under-approximation [13]. Set-valued and uncertainty-aware variants propagate zonotopes or augment the latent state with epistemic uncertainty [23, 29]. We do not claim the first latent safe set, collision observation, latent reachability filter, latent CBF, or transfer theorem. Our audits precede the downstream filter and ask which feasible decisions its input representation can still distinguish.

Safety-aware representations. Decision-sufficient and safe-RL representation learning already preserve action, reward, cost, feasibility, or safety features [5, 9, 14, 28, 30]. PIGDreamer additionally uses privileged representation alignment to improve safe partially observed world models [10]. Our target distinction is therefore not merely another auxiliary safety loss or privileged alignment method. The phrase *safe-action sufficiency* here denotes preservation of a nonempty common feasible action over every represented history, rather than the structural action-sufficient variables of Huang et al. [9]. The intervention directly optimizes a set-wise conflict object; profile prediction is an ablation, not by itself the novelty. We test it with observation-level controls, within-family utility retention, head removal, and a held-fixed downstream controller.

Stochastic encoders and testing. Variational autoencoders represent an input by a conditional law and regularize it toward a common prior; β -VAE changes the weight of that information bottleneck [8, 12]. Mutual singularity, total variation, binary testing, Pinsker’s inequality, and Markov-kernel data processing are classical. We do not claim new versions of those results. Our use of them is to state the exact almost-sure

safety semantics of a sampled code, expose the full-support Gaussian obstruction, and connect encoder overlap to unavoidable common-action violations. The experiments instead audit posterior means, so sampled-code and deterministic-mean conclusions remain separate.

Control abstractions and partial observation. Feedback refinement relations formalize controller refinement through static quantizers and state-dependent admissible inputs [21]; state-action abstractions likewise study which decisions remain representable after compression [2], while safety-preserving approximate bisimulation quantifies conservative abstraction error [7]. Under partial observation, belief-support shields already restrict actions to enforce almost-sure reach-avoid specifications [24]. Common-action necessity is therefore a foundation, not our standalone novelty. The submission-level target is approximate safety sufficiency for learned history representations, with calibrated audits and matched-utility intervention results.

History compression and approximate information states. Approximate information states provide learnable history compressions, latent dynamic programs, and policy-loss bounds [26]; worst-case variants treat uncertain partially observed systems and maximum-over-time costs [6]. Approximate Q -profile abstractions likewise give near-optimal policies under uniform value similarity [1]. Consequently, our finite-tree Bellman calculations only define and validate audit quantities; we do not present generic history compression, Hausdorff successor error, or factor-two policy loss as standalone novelty.

Calibration and robust planning. Conformalized latent constraints and robust MPC address model and constraint prediction errors [18]; uncertainty-aware reachability targets OOD failures [23], and partial-observation barrier-value methods combine estimator uncertainty with finite-horizon probabilistic safety [11]. These guarantees have different semantics from support-robust information loss inside an encoder fiber. The error sources are conceptually distinct but interacting; their end-to-end composition remains unresolved.

3 Safe-action sufficiency of representations

Let $x \in \mathcal{K}$ denote a physical state, history, or belief; $O(x)$ its observation; and $R(x) = E(O(x)) \in \mathcal{Z}$ the representation used for decisions. Let the safe set be $\mathcal{C} = \{x : h(x) \geq 0\}$. Cross-task statements use signed distance under a declared state metric; otherwise every defect is explicitly relative to the numerical margin h .

Definition 1 (Static fiber defect). *The exact h -relative separation defect is*

$$\varepsilon_{\text{sep}}(R) = \sup(\{h(x) : h(x) \geq 0, \exists x' \in \mathcal{K}, h(x') < 0, R(x') = R(x)\} \cup \{0\}).$$

For the implemented Euclidean audit, the radius- δ population object is

$$\varepsilon_{\text{sep}}^{\delta}(R) = \sup(\{h(x) : h(x) \geq 0, \exists x' \in \mathcal{K}, h(x') < 0, \|R(x) - R(x')\|_2 \leq \delta\} \cup \{0\}).$$

Other operational defects require an explicitly declared latent uncertainty set or metric.

For arbitrary latent subsets, the exact defect is intended to characterize the *infimum* completeness margin of a sound certificate. In general, impossibility is strict below the supremum and achievability is strict above it; endpoint and regularity conditions matter. For a stochastic encoder we instead write P_x for the output law and require a measurable accepted set to have probability zero under every unsafe law and probability one under every required safe law. This almost-sure semantics is deliberately stronger than a small expected or Bayes error. The experiments below audit deterministic posterior means, so their reported fibers must not be reinterpreted as audits of sampled posterior laws.

Static separation is insufficient for control. Let $\mathcal{U}_{\text{safe}}(x)$ contain actions satisfying a declared one-step, barrier, robust-MPC, or viability condition.

Definition 2 (Fiber-wise robust action set). *For the fiber $F_z = \{x : R(x) = z\}$, define*

$$\mathcal{U}_{\text{safe}}^R(z) = \bigcap_{x \in F_z} \mathcal{U}_{\text{safe}}(x).$$

A fiber is action-conflicted when each state is individually feasible under the declared condition, meaning $\mathcal{U}_{\text{safe}}(x)$ is nonempty, but $\mathcal{U}_{\text{safe}}^R(z)$ is empty.

An action-conflicted fiber is an immediate witness that no deterministic memoryless policy $\pi(z)$ satisfies the audited safety condition for every state in that fiber. Positive results will require nonempty intersections, dynamics/transition sufficiency, and measurable-selection or regularity assumptions.

For the empirical study, the condition is induced by a finite registered library $\mathcal{A}$ of actions, short open-loop sequences, or closed-loop skill primitives. Given a declared continuation rule and horizon H , define the counterfactual action profile and feasible set

$$q_H(x, a) = \min_{0 \leq t \leq H} h(x_t^{(a)}), \quad \Gamma_H(x) = \{a \in \mathcal{A} : q_H(x, a) \geq 0\}.$$

The notation does not denote a reward Q -function: it records the worst safety margin along the declared branch only. With multiple constraints h_r , every physical normalization scale $s_r > 0$ is frozen before outcomes are inspected and the joint profile is $q_H(x, a) = \min_{r, t \leq H} h_r(x_t^{(a)})/s_r$; the unaggregated constraint profiles are reported as secondary outcomes. For a viable reference set $\mathcal{V} = \{x : \Gamma_H(x) \neq \emptyset\}$, we call R *safe-action sufficient* for $(h, H, \mathcal{A})$ on $\mathcal{V}$ when

$$\bigcap_{x \in F_z \cap \mathcal{V}} \Gamma_H(x) \neq \emptyset \quad \text{for every } z \text{ with } F_z \cap \mathcal{V} \neq \emptyset.$$

This is deliberately specification-, horizon-, continuation-, and library-relative. It is not a claim of universal control sufficiency.

We write the deployed decision representation as

$$R_\theta(x) = A_\theta(E_\theta(O(x))),$$

where E_θ produces spatial or spatiotemporal tokens and A_θ includes any pooling, quantization, or rate-limited aggregation on which the controller acts. This location matters: post-processing a frozen R_0 cannot split its exact fibers. An intervention advertised as exact repair must therefore update modules that still access distinctions before the audited bottleneck, alter token assignment/quantization, or add new information. A deterministic head or adapter of the form $T \circ R_0$ can change finite-radius geometry but is a negative control for exact repair.

Finally, if $R_2 = T \circ R_1$, then every R_1 fiber is contained in an R_2 fiber. This data-processing relation makes the observation map a lower floor: deterministic encoding cannot recover safety information that O has already aliased. This exact comparison uses the same input domain and h -relative defect. A separate countable-state theorem gives the analogous mutual-singularity order for stochastic encoders and Markov-kernel post-processing. Neither result is an additive decomposition at finite radius or a comparison of one-frame observations with richer histories; finite-radius observation and state views require separately declared geometries.

4 Foundations for safe-action sufficiency

These results provide scoped audit primitives. Safe/unsafe aliasing, common admissible actions, and controller refinement are established ideas; submission-level novelty must come from a validated representation audit and learning intervention, not from the set-theoretic facts alone. The complete proofs and technical qualifications are in Appendix B. All suprema and infima are extended-real, with $\sup(\emptyset \cup \{0\}) = 0$ and $\inf \emptyset = +\infty$.

4.1 Static soundness and completeness

Let

$$\mathcal{C} = \{x \in \mathcal{K} : h(x) \geq 0\}, \quad \mathcal{D} = \{x \in \mathcal{K} : h(x) < 0\},$$

and let $R : \mathcal{K} \rightarrow \mathcal{Z}$ be deterministic. A latent set $S \subseteq \mathcal{Z}$ is *sound* if $R^{-1}(S) \subseteq \mathcal{C}$. For $m \geq 0$, it is closed m -complete when $\{x : h(x) \geq m\} \subseteq R^{-1}(S)$ and open m -complete when $\{x : h(x) > m\} \subseteq R^{-1}(S)$. Define

$$\Gamma_R = \{h(x) : h(x) \geq 0, \exists x' \in \mathcal{D} \text{ with } R(x') = R(x)\}, \quad \varepsilon_{\text{sep}}(R) = \sup(\Gamma_R \cup \{0\}).$$

Theorem 1 (Exact existence equivalence). *Let $m \geq 0$ and $S_{\max} = \mathcal{Z} \setminus R(\mathcal{D})$. A latent set S is sound if and only if $S \subseteq S_{\max}$. A sound closed m -complete set exists exactly when*

$$\Gamma_R \cap [m, +\infty) = \emptyset, \tag{1}$$

whereas a sound open m -complete set exists exactly when

$$\Gamma_R \cap (m, +\infty) = \emptyset. \tag{2}$$

Whenever the relevant condition holds, $S_{\max}$ attains it.

Proposition 1 (Infimum completeness threshold). *Let $\tau_{\geq}(R)$ and $\tau_{>}(R)$ be the infima over $m \geq 0$ for which a sound closed or open m -complete set exists, respectively. Then*

$$\tau_{\geq}(R) = \tau_{>}(R) = \varepsilon_{\text{sep}}(R).$$

For finite ε_{sep} , open completeness is attainable at $m = \varepsilon_{\text{sep}}$; closed completeness is attainable there if and only if $\varepsilon_{\text{sep}} \notin \Gamma_R$. If $\varepsilon_{\text{sep}} = +\infty$, no finite margin is attainable.

The endpoint distinction is substantive: a safe boundary point with $h = 0$ can share a code with an unsafe point, giving $\varepsilon_{\text{sep}} = 0$ while closed zero-completeness remains impossible. The maximal sound-set construction itself is already used by Lutkus et al. [13]; the contribution here is the exact threshold and endpoint accounting, not the collision observation.

4.2 Exact deterministic post-processing

Theorem 2 (Exact deterministic post-processing monotonicity). *If $R_2 = T \circ R_1$ for deterministic maps R_1, T , then*

$$\Gamma_{R_1} \subseteq \Gamma_{R_2}, \quad \varepsilon_{\text{sep}}(R_1) \leq \varepsilon_{\text{sep}}(R_2), \quad \tau_{\geq}(R_1) \leq \tau_{\geq}(R_2).$$

In particular, $\varepsilon_{\text{sep}}(O) \leq \varepsilon_{\text{sep}}(E \circ O)$ for a deterministic observation map and encoder.

This is an exact-fiber comparison on one input domain. A raw latent radius has no invariant law: an injective rescaling preserves fibers while changing neighborhoods. The result neither decomposes finite-radius ambiguity nor compares one frame with a richer history. Finite-radius observation and privileged-state controls require separately declared geometries.

4.3 Countable stochastic encoders

Let $\mathcal{K}$ be countable, let $(\mathcal{Z}, \Sigma)$ be an arbitrary measurable space, and let P_x be the encoder-output law at x . A measurable S is almost-surely sound when $P_x(S) = 0$ for every $x \in \mathcal{D}$ and is closed (open) m -complete when $P_x(S) = 1$ for every x with $h(x) \geq m$ (respectively $h(x) > m$). Write $P \perp Q$ for mutual singularity and define

$$\Gamma_P^{\text{st}} = \{h(x) : h(x) \geq 0, \exists x' \in \mathcal{D} \text{ with } P_x \not\perp P_{x'}\}, \quad \varepsilon_{\text{st}}(P) = \sup(\Gamma_P^{\text{st}} \cup \{0\}).$$

Theorem 3 (Almost-sure stochastic separation). *An almost-surely sound closed (open) m -complete separator exists exactly when Γ_P^{st} avoids $[m, +\infty)$ (respectively $(m, +\infty)$). Both infimum completeness thresholds equal $\varepsilon_{\text{st}}(P)$, with the same open/closed endpoint distinction as Proposition 1. When Σ separates realized codes (as in a standard Borel space), Dirac laws $P_x = \delta_{R(x)}$ recover the deterministic theory. If a Markov kernel K post-processes the code laws to $Q_x = P_x K$, then*

$$\Gamma_P^{\text{st}} \subseteq \Gamma_Q^{\text{st}}, \quad \varepsilon_{\text{st}}(P) \leq \varepsilon_{\text{st}}(Q).$$

For $\|P - Q\|_{\text{TV}} = \sup_{S \in \Sigma} |P(S) - Q(S)|$, the minimum unweighted sum of the two separator errors is exact and the same overlap also controls randomized latent actions.

Proposition 2 (Total-variation separation and action tradeoff). *For any safe-unsafe pair (x, x') ,*

$$\inf_{S \in \Sigma} \{P_x(S^c) + P_{x'}(S)\} = 1 - \|P_x - P_{x'}\|_{\text{TV}}.$$

If two states have disjoint measurable safe-action sets A_x, A_y and π is a randomized latent policy kernel, let $M_x = P_x \pi$, $M_y = P_y \pi$ and define $v_x = M_x(A_x^c)$, $v_y = M_y(A_y^c)$. Then

$$v_x + v_y \geq 1 - \|P_x - P_y\|_{\text{TV}}.$$

Thus, for these disjoint safe-action sets, simultaneous almost-sure action safety requires singular code laws. More generally, write (P, Q) for the relevant pair of code laws in either display. If both have finite KL to a common prior Π , Pinsker and the triangle inequality further lower-bound the corresponding error sum by

$$\left[1 - \sqrt{D_{\text{KL}}(P\|\Pi)/2} - \sqrt{D_{\text{KL}}(Q\|\Pi)/2}\right]_+.$$

Any two nondegenerate Gaussians on $\mathbb{R}^d$ are mutually absolutely continuous, so no nontrivial safe-unsafe pair admits exact almost-sure separation under such posterior laws. This does not describe our experiments, which audit the deterministic posterior mean rather than sampled posterior codes.

4.4 Common actions and finite-action slack

For a code fiber $F_z^R = \{x : R(x) = z\}$ and declared safe-action sets $\mathcal{U}_{\text{safe}}(x)$, define

$$\mathcal{A}_R(z) = \bigcap_{x \in F_z^R} \mathcal{U}_{\text{safe}}(x).$$

A fiber is action-conflicted when each member is individually feasible but $\mathcal{A}_R(z) = \emptyset$.

Theorem 4 (Deterministic action conflict). *A deterministic memoryless decision $\pi(z)$ satisfies the audited condition for every $x \in F_z^R$ if and only if $\pi(z) \in \mathcal{A}_R(z)$. Hence an action-conflicted fiber rules out every uniformly feasible deterministic memoryless latent decision.*

For a finite action library and real safety profile $q(x, a)$, let $\mathcal{V}_q = \{x : \max_a q(x, a) \geq 0\}$ be the individually feasible set and define the exact worst-fiber common-action violation on viable states

$$\beta_{\text{CA}}^{\mathcal{V}_q}(R) = \sup_{\substack{z \in R(\mathcal{K}) \\ F_z^R \cap \mathcal{V}_q \neq \emptyset}} \left[-\max_a \inf_{x \in F_z^R \cap \mathcal{V}_q} q(x, a) \right]_+.$$

Proposition 3 (Exact common-action data processing). *If $R_2 = T \circ R_1$ for deterministic maps, then $\beta_{\text{CA}}^{\mathcal{V}_q}(R_1) \leq \beta_{\text{CA}}^{\mathcal{V}_q}(R_2)$. Therefore deterministic post-processing of a frozen code cannot reduce its exact worst-fiber common-action violation, although a non-injective metric warp may alter a finite-radius audit.*

Randomization requires care. Under the stated measurability hypotheses, strong safety means placing probability one on $\mathcal{A}_R(z)$; pointwise almost-sure safety only places probability one on each individual $\mathcal{U}_{\text{safe}}(x)$. They are equivalent for countable fibers and for compact metric action spaces with closed safe-action sets, but not for arbitrary uncountable fibers (Theorem 7). When the common set is nonempty, Euclidean closed-convex action sets also admit a unique least-intervention projection of a nominal action (Theorem 8); this is pointwise in z and does not establish a measurable selector or recursive safety.

For continuous actions, suppose $p \geq 1$, $\mathcal{U} \subset \mathbb{R}^p$ is nonempty, compact, and convex, and every $\mathcal{U}_{\text{safe}}(x)$ in a fixed fiber is a nonempty closed-convex subset of $\mathcal{U}$. For nonempty G in that fiber set

$$\beta_{\text{geo}}(G) = \min_{u \in \mathcal{U}} \sup_{x \in G} d_2(u, \mathcal{U}_{\text{safe}}(x)).$$

Theorem 5 (Helly-compressed convex action audit). *For any nonempty, possibly infinite, realized fiber F_z^R ,*

$$\beta_{\text{geo}}(F_z^R) = \sup_{\substack{G \subseteq F_z^R \\ 1 \leq |G| \leq p+1}} \beta_{\text{geo}}(G).$$

The fiber has a common safe action exactly when every at-most- $(p+1)$ -state subfiber does. It is action-conflicted exactly when $\beta_{\text{geo}}(F_z^R) > 0$. For a finite fiber the supremum is a maximum, so at most $p+1$ states preserve the full geometric slack; the cardinality is sharp.

If $q = \dim \text{aff}(\mathcal{U}) < p$, the same statement holds with the sharper $q+1$ bound after passing to affine coordinates.

This is a fixed-fiber consequence of classical Helly theory. The norm-relative slack is not a failure probability or barrier margin, and the result supplies neither a measurable selector across codes nor recursive safety. Convexity and compactness are substantive; sampled or numerically approximate constraints remain finite-reference evidence unless independently verified.

For $z \in R(\mathcal{K})$, a finite action set $\{1, \dots, k\}$, and nonnegative violation loss with $\sup_{x \in F_z^R} \ell(x, a) < +\infty$ for every action, define

$$\beta_{\text{det}}(z) = \min_a \sup_{x \in F_z^R} \ell(x, a), \quad \beta_{\text{exp}}(z) = \inf_{p \in \Delta_k} \sup_{x \in F_z^R} \sum_{a=1}^k p_a \ell(x, a).$$

The state is worst-case after p is fixed, but loss is averaged over the sampled action. In contrast, the action-wise robust expected and worst-supported-action losses satisfy

$$\begin{aligned} \beta_{\text{ar}}(z) &= \inf_{p \in \Delta_k} \sum_a p_a \sup_{x \in F_z^R} \ell(x, a), \\ \beta_{\text{ws}}(z) &= \inf_{p \in \Delta_k} \max_{a: p_a > 0} \sup_{x \in F_z^R} \ell(x, a). \end{aligned}$$

Theorem 6 (Finite-action pointwise expected-loss slack). *For every nonnegative loss,*

$$\frac{1}{k} \beta_{\text{det}}(z) \leq \beta_{\text{exp}}(z) \leq \beta_{\text{det}}(z), \quad \beta_{\text{ar}}(z) = \beta_{\text{ws}}(z) = \beta_{\text{det}}(z). \quad (3)$$

If $\ell(x, a) = 0$ exactly for safe actions and the fiber is action-conflicted, all four slacks are strictly positive. For hard failure loss, $\beta_{\text{det}} = \beta_{\text{ar}} = \beta_{\text{ws}} = 1$ and $\beta_{\text{exp}} \geq 1/k$. A finite-fiber linear-program dual supplies a traceable adversarial state mixture; see Appendix B.3.

4.5 Average world-model loss does not control a worst fiber

Proposition 4 (Smooth localized collision). *Fix $a \in (0, 1)$, let X be uniform on $[-1, 1]$, set $h(x) = x$, and use identity dynamics $X^+ = X$. For every $0 < \delta < \min\{a, 1-a\}$ there are smooth scalar encoder, decoder, and latent predictor with*

$$\varepsilon_{\text{sep}}(R_\delta) \geq a, \quad \mathbb{E}|X - D(R_\delta(X))|^2 \leq 2a^2\delta, \quad P(R_\delta(x)) = R_\delta(x), \quad \mathbb{E}|X^+ - D(P(R_\delta(X)))|^2 \leq 2a^2\delta.$$

The same collided fiber is action-conflicted for actions $\{-1, +1\}$ and condition $ux \geq 0$.

The construction modifies the identity encoder only inside two shrinking smooth bump supports, while mapping a and $-a$ to the same code. Thus average reconstruction and decoded prediction loss vanish even though the worst static and action obstruction stays fixed. A positive implication requires tail coverage, uniform regularity, or verification; a small empirical mean is insufficient.

All quantities remain specification-relative: replacing h by ch scales the defect, and encoding one known scalar margin does not preserve dynamics or common actions. The static frontier optimizes over arbitrary latent subsets rather than a neural or verifiable certificate class. Continuous nonconvex actions, approximate neighborhoods, uncountable stochastic state families, recursive feasibility, and long-horizon safety require additional assumptions not supplied here.

5 Auditing and learning

Finite-sample witnesses. Let $z_i = R(x_i)$, $h_i = h(x_i)$, and let $q_i(a)$ be the declared finite-action safety margin, with $q_i(a) \geq 0$ denoting feasibility of action a at sample i . At a frozen radius δ , the static finite-reference witness is

$$\hat{\varepsilon}_\delta = \max(\{h_i : h_i \geq 0, \exists j \text{ with } h_j < 0, \|z_i - z_j\|_2 \leq \delta\} \cup \{0\}).$$

Every returned pair is traceable to sample identifiers. Conditional on exact margins and distances, an observed pair lower-bounds the corresponding population radius supremum; absence of a pair supplies no population upper bound. An unthresholded nearest unsafe neighbor does not witness an exact collision and is retained only as a sampling-dependent diagnostic.

For action sufficiency, first restrict the reference population to individually viable indices $\mathcal{I}_0 = \{j : \max_a q_j(a) \geq 0\}$. For each $i \in \mathcal{I}_0$, center the viable neighborhood $N_i^\mathcal{V}(\delta) = \{j \in \mathcal{I}_0 : \|z_j - z_i\|_2 \leq \delta\}$ and retain it only when $|N_i^\mathcal{V}(\delta)| \geq 2$. Its required common-action violation is

$$v_i(\delta) = \left[-\max_a \min_{j \in N_i^\mathcal{V}(\delta)} q_j(a) \right]_+.$$

This detects conflicts involving more than two states and records maximizing witnesses, eligible denominator, and conflict fraction; a trajectory with no eligible center fails rather than scoring zero. The primary summary is the arithmetic mean across trajectories of their linearly interpolated 0.95 quantiles; seeds, not trajectories, are paired for inference. A repair weight is eligible only if every validation stratum satisfies the preregistered within-family native-utility margin, retains the required center and trajectory coverage, and preserves the improvement sign under a radius chosen to match the frozen encoder’s nonself neighborhood mass. Exact gates, radii, and multiplicity corrections freeze in the foundation-study protocol before test access. Trajectories are resampled only within seed for tail estimation. The global maximum remains a lower witness, not the sole endpoint. Continuous-action claims require verified, not grid-only, optimization.

Operational calibration. Raw latent radii change under code rescaling. A calibration split may define an exploratory scale convention; robust or certified semantics require a deployment-grounded uncertainty set or verified set propagation. We retain translation, rotation, rescaling, and irrelevant-coordinate controls. A radius expressed as a fraction of calibration-set median pairwise distance is only a scale-normalized exploratory view; the conclusion must also survive a frozen radius curve, matched-neighborhood-mass analysis, and physically generated perturbation or alias pairs. Exact-code synthetic tests serve as unit tests, not evidence that continuous neural encoders collide in practice.

Buffered action profiles. Let $\mathcal{A} = \{1, \dots, K\}$ be the registered action or skill library and let $q_i(a)$ be the margin produced by the declared rollout from history x_i . For a buffer $\gamma \geq 0$, retain the individually feasible reference set

$$\mathcal{I}_\gamma = \{i : \max_{a \in \mathcal{A}} q_i(a) \geq \gamma\}.$$

For nonempty $B \subseteq \mathcal{I}_\gamma$, its common-action margin and violation are

$$c_\gamma(B) = \max_{a \in \mathcal{A}} \min_{j \in B} \{q_j(a) - \gamma\}, \quad v_\gamma(B) = [-c_\gamma(B)]_+.$$

Thus $v_\gamma(B) = 0$ exactly when one declared action has margin at least γ for every member. The audit above is the case $\gamma = 0$ and $B = N_i^\mathcal{V}(\delta)$. Profiles may come from exhaustive simulator branching, a verified model, or a cross-fitted action-conditioned teacher, but their source and error semantics are reported separately. If a finite-reference teacher has a uniform error bound η , both c_γ and v_γ change by at most η ; a percentile error gate does not imply this uniform statement.

Common-Action Geometry Repair. Let R_0 be a pretrained representation and let R_θ update modules that still access tokens or observations before the audited bottleneck. Write $z_i^\theta = R_\theta(x_i)$ and let $d_\theta(i, j)$ be Euclidean distance after a calibration-frozen normalization. For $\delta_{\text{train}} > \delta_{\text{audit}}$, define

$$C_i^\theta(a) = \sum_{j \in \mathcal{I}_\gamma} [\gamma - q_j(a)]_+ [\delta_{\text{train}} - d_\theta(i, j)]_+^2, \quad \mathcal{L}_{\text{CA}}(\theta) = \frac{1}{|\mathcal{I}_\gamma|} \sum_{i \in \mathcal{I}_\gamma} \min_{a \in \mathcal{A}} C_i^\theta(a).$$

The action minimum is replaced by a temperature-controlled soft minimum for optimization. On a finite reference set, a zero hard loss implies that every strict δ_{train} neighborhood restricted to $\mathcal{I}_\gamma$ has a common action with margin at least γ ; sampled minibatches or mined neighborhoods do not promote this statement to the population. Unlike a pairwise contrastive loss, the action minimum is shared by the entire centered neighborhood and can therefore expose higher-order conflicts whose safe-action sets intersect pairwise but not jointly.

An action-conditioned head $\hat{q}_{\theta, \psi}(x, a) = g_\psi(R_\theta(x), a)$ supplies a tail-sensitive, boundary-weighted profile loss $\mathcal{L}_{\text{prof}}$. The complete objective is

$$\mathcal{L}_{\text{repair}} = \mathcal{L}_{\text{native}} + \lambda_{\text{prof}} \mathcal{L}_{\text{prof}} + \lambda_{\text{CA}} \mathcal{L}_{\text{CA}} + \lambda_{\text{retain}} \mathcal{L}_{\text{retain}}.$$

Here $\mathcal{L}_{\text{native}}$ retains the model’s original utility and $\mathcal{L}_{\text{retain}}$ anchors frozen-teacher features or calibration-set pairwise geometry. These terms are family-relative rather than a single commensurate score. Cosmos retains reconstruction/perceptual and rate utility through its decoder; V-JEPA retains latent prediction and frozen-teacher features; DINO retains dense frozen-teacher feature geometry [4, 20, 25].

Where repair occurs. For continuous Cosmos, the primary intervention places residual bottlenecks before the tokenizer latent and evaluates reconstruction through a frozen decoder. For V-JEPA and DINO, low-rank or bottleneck modules enter the final internal transformer blocks before the deployed pooling and rate constraint. A deterministic post-code adapter $T_\theta \circ R_0$ cannot split exact R_0 fibers by data processing; it is retained only as a finite-radius geometry control. A fixed discrete-token remapping has the same limitation, so discrete-token repair must update the encoder or token assignment. Appending the true or predicted profile is a side-channel upper bound, not evidence that the original representation was repaired.

Separating representation repair from probe fitting. After post-training, we discard the joint profile head, freeze R_θ , recompute the nonparametric defect from held-out physical profiles, and fit a fresh capacity-matched readout and downstream selector on a disjoint split. The decisive controls are a frozen encoder with the same head, profile-only and common-action-only internal adaptation, pairwise conflict separation, native- loss-only adaptation, post-code adaptation, scalar-margin and FCSRL-style supervision, shuffled profiles, conflict-count-matched random groups, and an oracle appended profile. Unit-normalized codes, frozen radius curves, matched-neighborhood-mass or fixed- k audits, physical perturbation groups, effective rank, distance spectra, retained viable coverage, and token entropy prevent a trivial win by uniformly spreading codes or emptying neighborhoods.

6 Controlled evidence and prospective evaluation

6.1 Registered finite dynamic controls

We first isolate partial observability without training an encoder. For each controlled domain, two hand-selected roots have the same current 32×32 rendered frame and different hidden velocity. From each root we exhaustively enumerate all 3^3 sequences on the registered action grid $\{-1, 0, +1\}$ under nominal deterministic dynamics. Let $V_s(\eta)$ be the full-history max-min margin-to-go and $Q_s(\eta, a)$ the value after fixing the first action and then allowing full-history optimal control to resume. On the viable members of a code fiber, the first-action obstruction is

$$\kappa_s(z) = \min_a \max_{\eta: R_s(\eta)=z, V_s(\eta) \geq 0} [-Q_s(\eta, a)]_+.$$

Thus $\kappa_s(z) = 0$ exactly when the viable histories in that finite fiber share a viable first action; empty viable fibers use zero. A separate backward-induction pass constructs a time-indexed common-safe-action policy

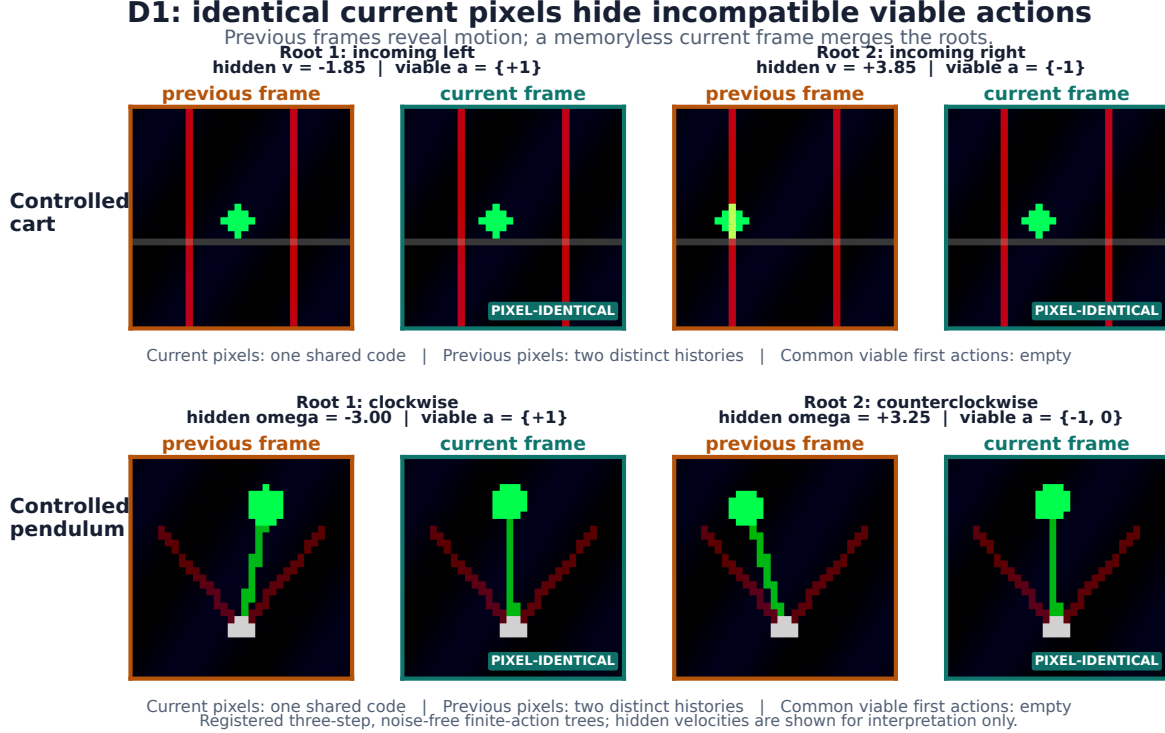


Figure 1: Registered D1 hidden-velocity aliases. Within each domain, the two roots have pixel-identical current frames but incompatible viable first-action sets; the preceding frame separates the histories. Images and action sets are exhaustive only for the declared three-step, noise-free nominal trees. Hidden-state labels are explanatory and are not inputs to the memoryless code.

and checks retained viability throughout the tree. Global-policy signs use exact comparisons on stored binary floats; 10^{-12} applies only to numerical-bound checks and the separately labeled $\kappa_s \leq 10^{-12}$ diagnostic. Here $h_{\text{cart}} = 0.82 - |p|$ and $h_{\text{pend}} = 0.70 - |\text{wrap}(\theta)|$.

Table 1: Exact-code controls on the registered three-step nominal trees. *Global policy* means one representation policy preserves every full-history-viable history on all enumerated layers.

Domain	representation code	root κ_3 (native)	global policy
Cart	privileged state	0	yes
Cart	current frame	0.017987	no
Cart	causal two-frame history	0	yes
Pendulum	privileged state	0	yes
Pendulum	current frame	0.015351	no
Pendulum	causal two-frame history	0	yes

At the cart roots the individually viable action sets are respectively $\{+1\}$ and $\{-1\}$; at the pendulum roots they are $\{+1\}$ and $\{-1, 0\}$. The exact current-frame code therefore admits no common viable action in either domain, while the previous frame separates the roots. Figure 1 visualizes the registered alias construction, and Table 1 reports its exact finite-tree audit. Both are provisionally internally verified finite floating-point fixtures, not learned-model or population evidence; their scope is the declared roots, actions, horizon, arithmetic semantics, and noise-free model. The cart roots also lie outside the behavior generator’s nominal velocity envelope. Exact roots, configuration, and the producing command are registered under D1-controlled-dynamic-oracle in the repository claim ledger.

6.2 Prospective pretrained-representation study

No pretrained-encoder result is reported. The checked-in AE/ β -VAE cart, pendulum, and Dubins pipeline remains an engineering precursor for estimator, profile-label, and confirmatory-analysis plumbing; it is not the headline empirical study and supplies no foundation-model evidence. The new protocol addresses three questions: whether standard pretrained features exhibit safe-choice aliasing, whether the proposed audit predicts closed-loop failures beyond native utility and a safety probe, and whether an internal encoder repair improves both quantities without erasing the original representation utility.

Pretrained families and information matching. The primary frozen checkpoints deliberately span three objectives. Cosmos Tokenizer supplies continuous and discrete causal video codes at controlled spatial and temporal compression rates [19, 20]. V-JEPA 2.1 ViT-B supplies dense temporal predictive features, with V-JEPA 2-AC as an action-conditioned sensitivity [4, 15]. DINOv3 ViT-S/B supplies semantic and dense spatial features [25]. Every model receives the same causal RGB clip, time span, and available proprioceptive variables. Because DINO is image-based, its patch features are encoded framewise and passed through the same capacity-matched temporal aggregator used in the standardized comparison. We report both each model’s native interface and a matched-rate interface with the same token/bit budget and downstream pooling capacity; full token grids and pooled decision states are audited separately. Continuous features use a registered quantizer and numerical precision; floating-point dimension alone is not treated as bitrate. Cross-family results are descriptive because architecture, scale, and pretraining data differ. A smaller matched-backbone, matched-data objective ablation is required before attributing differences causally to the pretraining objective.

Two primary physical domains. The first target is contact-rich manipulation in ManiSkill3 [27]. A finite library contains short Cartesian end-effector motions and gripper skills; profiles record minimum collision-clearance, joint-limit, contact-force, workspace, and object-drop margins. The second target is embodied navigation in Habitat [22]. Its library contains short velocity or waypoint skills; profiles record minimum obstacle clearance, stopping margin, and, when present, distance to moving agents. The two domains differ in appearance, dynamics, contact, and action semantics. Driving may be added as a third external-validity study, but no driving-specific model is required for the core claim.

Each example begins from a saved simulator state. The encoder receives observations only, while the data generator branches that state under every registered skill to produce $q_H(x, \cdot)$. This privileged counterfactual oracle is the primary mechanism experiment, not a deployment claim. A cross-fitted action-conditioned teacher trained without held-out simulator state is a realism ablation and must report profile error and abstain near the safety boundary. Raw causal video and privileged physical state provide information upper bounds; one-frame observations and exact aliases isolate the observation floor.

Repair and baseline arms. Within each pretrained family, data, history, trainable parameter count, optimizer steps, and readout capacity are paired. The required arms are: frozen encoder plus profile head; native-loss- only internal adaptation; scalar-margin and frozen FCSRL-style feasibility adaptation [5]; internal profile-only adaptation; pairwise conflict separation; the full set-wise repair; and a post-code adapter explicitly labeled geometry-only. Shuffled profiles, conflict-count-matched random groups, true-profile append, and full fine-tuning are controls. The same profile head is removed from every learned arm before a fresh head and controller are trained. Recent planning-aware driving tokenization already supplements reconstruction with representation and geometry signals [31]; accordingly, a generic auxiliary-loss improvement is not the claimed contribution.

Endpoints and held-fixed downstream test. The primary representation endpoint is the trajectory-balanced tail of $v_i(\delta)$, normalized by physical margin scale and reported as a frozen radius curve, at matched neighborhood mass or fixed k , and on physically generated groups. Coverage, duplicate rate, effective rank, distance distortion, and token entropy accompany every value. Family-specific native utility is assessed against its own frozen checkpoint—reconstruction/perceptual/rate metrics for Cosmos, latent prediction for V-JEPA, and frozen dense-feature transfer for DINO—and is not averaged across families.

For downstream evaluation, one frozen nominal controller proposes or ranks the same registered skills in each condition. A freshly trained capacity-matched safety readout filters or reranks them, after which controller parameters remain fixed. Primary outcomes are constraint violations and near misses; task success, return, intervention rate, and control effort are joint utility outcomes. We test whether the common-action audit adds leave-one-domain-out predictive value beyond native utility, ordinary action prediction, and scalar-safety probe metrics. Appearance shifts, unseen hazard mechanisms, safety thresholds, profile horizons, and denser action libraries test specification and distribution transfer.

Advancement and falsification gates. All repair comparisons use fresh paired seeds and a validation-frozen utility noninferiority margin before test access. The foundation study advances only if raw causal observations predict the registered profiles, the defect survives information/rate matching in both domains and at least two families, the full internal repair beats an identical frozen-head and native-adaptor control, and the improvement survives head removal, matched-neighborhood-mass auditing, and a fresh downstream controller. The thesis is rejected or narrowed if observation ambiguity explains the effect, a head-only model matches repair, representation gains come only from norm expansion or empty neighborhoods, scalar feasibility dominates, or the audit adds no information about closed-loop violations.

7 Limitations and broader impact

Fiber verification can be as hard as neural verification; an order-statistic wrapper still needs the violation oracle. Static separation gives neither Markov dynamics nor long-horizon invariance. Learned specifications require error composition; transfer and operational radii remain deployment choices. The stochastic separator theorem uses countably many physical states and an almost-sure criterion; pairwise singularity need not combine into one separator for an uncountable family. Its Gaussian consequence concerns sampled full-support posterior laws, whereas our experiments use posterior means. The continuous-action compression requires exact nonempty closed-convex safe sets inside a compact convex actuator set; its action-space distance is not a calibrated violation cost, and numerical or sampled feasibility checks are not population certificates.

Safe-action sufficiency is relative to the registered safety margin, horizon, continuation rule, and finite action or skill library. A sampled library cannot support a claim for a continuous action space without verified inner optimization. Simulator branching supplies privileged counterfactual labels unavailable in many real systems; predicted teachers add model error, and a percentile validation gate is not a uniform certificate. Empirical neighborhoods approximate fibers only under a declared geometry and coverage regime and can be gamed by representation rescaling, dispersion, or pooling. Matched-neighborhood-mass, physical groups, retention metrics, and fresh-head tests diagnose but do not eliminate these limitations.

Cosmos, V-JEPA, and DINO differ in architecture, parameter count, data, input interface, and native objective. Their cross-checkpoint comparison is descriptive, while repair conclusions are paired within family; causal claims about pretraining objectives require the proposed matched-backbone and matched-data ablation. Native reconstruction, prediction, and dense-feature metrics are also not directly commensurate. The primary domains are simulated manipulation and navigation, so even a positive result would not establish real-robot, driving, or specification-universal safety.

We report no completed pretrained-encoder audit, internal repair, confirmatory comparison, foundation-scale downstream result, or population certificate. The existing CPU profile, FCSRL, world-model, and Dubins paths establish precursor plumbing only. Dynamic controls use hand-selected roots and known-dynamics profiles. Audits are not guarantees; releases will record witnesses, failed controls, coverage, and uncertainty assumptions.

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A Reproducibility and computational details

Every included theorem, proposition, table, and figure is registered in `claims.toml`. Validation rejects label drift and requires a producing command, immutable manifest, and reviewer; independent promotion additionally requires a collaborator rerun.

Executable artifact map. Notebook outputs are explanatory; ledger proof sources, manifests, and commands are authoritative.

Object	Reproduction entry point and authoritative artifact	Status
Included theory	Claim/label validation and finite regressions; TeX proof appendix and claim ledger	Draft; review pending
D1	Dynamic-oracle manifest check and exhaustive unit tests; registered evidence JSON	Provisional internal control
E0 controls	<code>scripts/build_notebooks.py</code> ; fresh-kernel execution; notebook 00 validator	Excluded checks
Legacy path	AE/ β -VAE config/profile runners, radius reduction, aggregation, and notebooks 06–07	Plumbing only
Foundation	Pretrained-checkpoint card, simulator branch manifest, repair configuration, fresh-head and downstream manifests	Prospective; no result

D1 finite-control methods card. The cart roots are $(p, v) = (-.30, -1.85)$ and $(-.30, 3.85)$; the pendulum roots are $(\theta, \omega) = (-.04, -3.00)$ and $(-.04, 3.25)$. In each domain they have algebraically reachable safe predecessors, equal current rendered frames, and distinct preceding frames. From each root the oracle enumerates every sequence in $\{-1, 0, +1\}^3$ with zero process noise under the shipped dynamics. State, current-frame, and two-frame codes use exact stored-tuple equality. Viability, common-action membership, and the global-policy Boolean use exact stored-float signs; 10^{-12} is confined to numerical-bound checks and the separately labeled near-zero obstruction diagnostic. The authoritative manifest is `evidence/controlled_dynamic_oracle_v1.json`; it records the predecessor states, complete config hashes, all table cells, and the non-population scope. The executable entry point is `scripts/run_dynamic_oracle.py`; the D1 ledger entry pins its manifest-check invocation and regression tests.

A.1 Frozen FCSRL feasibility-loss adaptation

This prospective same-backbone offline arm is an adaptation, not a reproduction of FCSRL’s online TD3–Lagrangian agent. It fixes data, encoder and transition architectures, reconstruction/ transition/decoder/KL losses, latent width, split, optimizer budget, and checkpoint rule. Let $s(y) = \text{sign}(y) \log(1 + |y|)$. A head $p_{\theta,j}(z)$ uses 63 equally spaced atoms $b_j \in [-2, 4]$ in s -space and mean $F_{\theta}(z) = s^{-1}(\sum_j b_j p_{\theta,j}(z))$.

For a 10-transition behavior-policy window, compute stop-gradient targets backward with

$$Y_9 = \bar{F}_{\theta}(H_{10}), \quad Y_t = \max\{\mathbf{1}\{h(x_t) < 0\}, .9(1 - d_t)[(1 - u_t)Y_{t+1} + u_t\bar{F}_{\theta}(H_{t+1})]\}, \quad t = 8:0,$$

where d_t is true termination, u_t is time-limit truncation, and $\bar{F}_{\theta}(H) = F_{\theta}(\bar{g}(H))$. The target head is stop-gradient but not EMA; after each optimizer step, $\bar{\theta}_g \leftarrow .99\bar{\theta}_g + .01\theta_g$. The first $K = 4$ unrolled predictions minimize cross-entropy against the piecewise-linear projection of $s(Y_t)$ onto the two adjacent atoms. The ending transition is retained and positions strictly after the first termination or truncation are masked. To match the base-example and optimizer-step exposure of the other arms, the implementation starts one sequence at each original transition; near the horizon it repeats the terminal history and a fixed padding action only after the retained truncation, where the mask is zero.

This implementation remains a comparator for the legacy AE/ β -VAE path. In the foundation study, only its categorical feasibility target is ported to each same pretrained backbone, with the same internal-adapter placement, data, parameter budget, and fresh-head evaluation as the proposed repair; the port is still an adaptation rather than an official reproduction [5]. No FCSRL target-latent cosine loss, actor, critic, or Lagrangian update is transplanted. It therefore estimates behavior-policy future violation, not the robust counterfactual profile; the pinned legacy crosswalk is `docs/fcsrl_baseline_adaptation.md`.

A.2 Prospective foundation-study reporting card

Every audited checkpoint must record its exact version and weight hash; license; preprocessing; input resolution, frames per second, and causal history span; selected layer and token field; pooling or quantization; native and matched-rate dimensions; frozen and trainable parameter counts; optimizer steps; peak memory; and wall-clock compute. The simulator manifest records the saved physical state, observation hash, action/skill definition, continuation rule, horizon, safety margin, all counterfactual branches, and oracle version for every example. Split manifests group correlated trajectories before assigning train, calibration, validation, and test.

The evaluation manifest separately pins native-interface and standardized-interface audits, normalization statistics, radius curves, matched-mass or fixed- k neighborhoods, physical groups, coverage, feature-rank and distance diagnostics, profile-head removal, fresh-readout training, and the held-fixed downstream controller. Model and radius selection use validation data only. A pretrained checkpoint comparison is reported as observational; only paired within-family adaptation supports a repair-effect claim, and even that claim is conditional on the registered simulator and action library.

B Proofs for the exact audit primitives

B.1 Static frontier and post-processing

Proof of Theorem 1. If a sound S contained $z \in R(\mathcal{D})$, an unsafe preimage of z would belong to $R^{-1}(S)$, a contradiction. Conversely, a preimage of a point outside $R(\mathcal{D})$ cannot be unsafe, so every subset of $S_{\max}$ is sound. Since every sound set is contained in $S_{\max}$, closed m -completeness is possible precisely when no x with $h(x) \geq m$ has a code in $R(\mathcal{D})$, which is Eq. (1). Replacing $\geq m$ by $> m$ gives Eq. (2). $\square$

Proof of Proposition 1. For every finite $m > \varepsilon_{\text{sep}}$, Eq. (1) holds. For $0 \leq m < \varepsilon_{\text{sep}}$, the definition of supremum gives an element of Γ_R strictly larger than m , so neither completeness condition holds. At a finite endpoint no element exceeds ε_{sep} , proving open attainability, while closed attainability holds exactly when the supremum is not itself in Γ_R . If the supremum is infinite, Γ_R is unbounded. Taking infima proves the result. $\square$

The endpoint can fail in both directions one might otherwise conflate. If a safe boundary state with $h = 0$ aliases an unsafe state, then $\varepsilon_{\text{sep}} = 0$ but closed zero-completeness is impossible. If conflicting margins approach one from below without attaining one, closed one-completeness is attainable even though $\varepsilon_{\text{sep}} = 1$.

Proof of Theorem 2. Equality under R_1 implies equality after applying T , so every conflicting pair for R_1 is also conflicting for R_2 . Hence $\Gamma_{R_1} \subseteq \Gamma_{R_2}$; the supremum inequality and Proposition 1 give the remaining conclusions. $\square$

B.2 Stochastic separation and post-processing

Proof of Theorem 3. Fix either the closed or open convention and let H_m be its required safe-state set. Necessity is pairwise: if $P_x(S) = 1$ for $x \in H_m$ and $P_{x'}(S) = 0$ for $x' \in \mathcal{D}$, then $P_x \perp P_{x'}$. Conversely, assume every such pair is singular. For each pair choose $S_{x,x'} \in \Sigma$ with $P_x(S_{x,x'}) = 1$ and $P_{x'}(S_{x,x'}) = 0$. Countability permits

$$S_x = \bigcap_{x' \in \mathcal{D}} S_{x,x'}, \quad S = \bigcup_{x \in H_m} S_x.$$

Empty intersections are $\mathcal{Z}$ and empty unions are empty. Countable intersection preserves $P_x(S_x) = 1$, while for every unsafe x' , countable subadditivity gives $P_{x'}(S) = 0$. Thus S is the common separator. The condition is exactly avoidance of the appropriate upper ray by Γ_P^{st} , so the supremum and endpoint argument in Proposition 1 applies verbatim.

If Σ separates realized points, then for Dirac laws $\delta_z \perp \delta_{z'}$ exactly when $z \neq z'$, giving $\Gamma_P^{\text{st}} = \Gamma_R$. Without that qualification two distinct but measurably indistinguishable codes induce the same law. Finally, total variation contracts through every Markov kernel. Indeed, for a target event B , set $f(z) = K(z, B) \in [0, 1]$ and use the layer-cake identity to obtain

$$\left| \int f dP_x - \int f dP_{x'} \right| \leq \int_0^1 |P_x(f > t) - P_{x'}(f > t)| dt \leq \|P_x - P_{x'}\|_{\text{TV}}.$$

Taking the supremum over B gives

$$\|P_x K - P_{x'} K\|_{\text{TV}} \leq \|P_x - P_{x'}\|_{\text{TV}}.$$

Probability laws are mutually singular exactly when their total variation is one. Hence a non-singular input pair cannot become singular after post-processing, proving $\Gamma_P^{\text{st}} \subseteq \Gamma_Q^{\text{st}}$ and the defect inequality. $\square$

Countability cannot simply be deleted. On $\mathcal{Z} = [0, 1]$, take one required safe state with law δ_t for every $t \in [0, 1]$ and one unsafe state with the uniform law. Every safe law is pairwise singular to the unsafe law, but a separator accepted almost surely by every δ_t must equal all of $[0, 1]$ and therefore has unsafe probability one.

Proof of Proposition 2. For any measurable S ,

$$P_x(S^c) + P_{x'}(S) = 1 - \{P_x(S) - P_{x'}(S)\}.$$

The Hahn decomposition of the finite signed measure $P_x - P_{x'}$ supplies a measurable set attaining the bracketed one-sided supremum, which equals total variation for probability laws. This proves the first identity and its attainability.

For the action statement, let $M_x = P_x \pi$ and $M_y = P_y \pi$ be the induced action laws. Since $A_x \cap A_y = \emptyset$,

$$M_x(A_x) - M_y(A_x) \geq (1 - v_x) - v_y = 1 - v_x - v_y.$$

Total-variation contraction through the policy kernel gives $1 - v_x - v_y \leq \|M_x - M_y\|_{\text{TV}} \leq \|P_x - P_y\|_{\text{TV}}$. For either relevant pair of code laws (P, Q) and a common prior Π , the triangle inequality and Pinsker's inequality give

$$\|P - Q\|_{\text{TV}} \leq \|P - \Pi\|_{\text{TV}} + \|Q - \Pi\|_{\text{TV}} \leq \sqrt{D_{\text{KL}}(P\|\Pi)/2} + \sqrt{D_{\text{KL}}(Q\|\Pi)/2}.$$

Substitution into the exact separator identity or the action inequality, followed by clipping a possibly negative lower bound at zero, proves the two KL statements. $\square$

Every Gaussian law on $\mathbb{R}^d$ with positive-definite covariance has a strictly positive Lebesgue density. Any two such laws are therefore mutually absolutely continuous, and in particular are not singular. Degenerate Gaussians can instead be singular on distinct affine supports, so nondegeneracy is essential.

B.3 Common-action and randomization details

Proof of Theorem 4. The single action $\pi(z)$ satisfies every compatible state's condition exactly when it belongs to their safe-action intersection. $\square$

Proof of Proposition 3. For every $z_1 \in R_1(\mathcal{K})$, its fiber $F_{z_1}^{R_1}$ is contained in the R_2 fiber $F_{T(z_1)}^{R_2}$, and intersecting both with the representation-independent set $\mathcal{V}_q$ preserves this inclusion. Hence, for every action a and every nonempty viable R_1 subfiber,

$$\inf_{x \in F_{T(z_1)}^{R_2} \cap \mathcal{V}_q} q(x, a) \leq \inf_{x \in F_{z_1}^{R_1} \cap \mathcal{V}_q} q(x, a).$$

Taking the maximum over actions, negating, applying the positive part, and finally taking the supremum over z_1 gives $\beta_{\text{CA}}^{\mathcal{V}_q}(R_1) \leq \beta_{\text{CA}}^{\mathcal{V}_q}(R_2)$. $\square$

Let $(\mathcal{U}, \Sigma_{\mathcal{U}})$ be a measurable action space and assume every $\mathcal{U}_{\text{safe}}(x)$ and $\mathcal{A}_R(z)$ is measurable. A randomized decision μ_z is *strongly safe* when $\mu_z(\mathcal{A}_R(z)) = 1$ and pointwise almost surely safe when $\mu_z(\mathcal{U}_{\text{safe}}(x)) = 1$ for every fixed x in the fiber.

Theorem 7 (Qualified randomized action conflict). *A strongly safe randomized decision exists if and only if $\mathcal{A}_R(z)$ is nonempty. The same equivalence holds under pointwise almost-sure safety if either the fiber is countable, or the action space is compact metric, μ_z is Borel, and every $\mathcal{U}_{\text{safe}}(x)$ is closed.*

Proof. A probability measure cannot put mass one on an empty common set, while a common action induces a safe Dirac measure. For a countable fiber, the intersection of the probability-one safe-action events still has probability one. In the compact-closed case every finite intersection has probability one and is nonempty. The closed sets therefore have the finite-intersection property, so compactness makes their full intersection nonempty. The converse is again a Dirac measure. $\square$

The qualification is necessary. Let the action space and an uncountable fiber both be indexed by $[0, 1]$, and set $\mathcal{U}_{\text{safe}}(x) = [0, 1] \setminus \{x\}$. The common set is empty, but the uniform action distribution is pointwise almost surely safe for every fixed state; it is not strongly safe.

Theorem 8 (Common-action projection). *Suppose actions lie in $\mathbb{R}^p$ and every $\mathcal{U}_{\text{safe}}(x)$ in a fixed fiber is closed and convex. If $\mathcal{A}_R(z)$ is nonempty, then for every nominal action $u_{\text{nom}}(z)$,*

$$\pi_{\text{LI}}(z) = \arg \min_{u \in \mathcal{A}_R(z)} \frac{1}{2} \|u - u_{\text{nom}}(z)\|_2^2$$

exists uniquely, is uniformly feasible on the fiber, and satisfies

$$\langle u_{\text{nom}}(z) - \pi_{\text{LI}}(z), u - \pi_{\text{LI}}(z) \rangle \leq 0 \quad \forall u \in \mathcal{A}_R(z).$$

Proof. The common set is a closed convex intersection. A minimizing sequence has bounded objective and is therefore bounded; in finite dimensions it has a convergent subsequence whose limit is feasible and minimizing. If two distinct minimizers u, v existed, their feasible midpoint would satisfy

$$\left\| \frac{u+v}{2} - u_{\text{nom}} \right\|^2 = \frac{1}{2} \|u - u_{\text{nom}}\|^2 + \frac{1}{2} \|v - u_{\text{nom}}\|^2 - \frac{1}{4} \|u - v\|^2,$$

a strict improvement. The variational inequality follows from the right derivative along the feasible segment from the minimizer to any u in the common set. $\square$

This result does not ensure that $z \mapsto \pi_{\text{LI}}(z)$ is measurable or continuous, nor that an infinite constraint intersection has a finite verified reduction. Barrier hypotheses needed for forward invariance remain separate.

Proof of Theorem 5. For $r \geq 0$ and $x \in F_z^R$, define the relative expansion

$$B_x(r) = \{u \in \mathcal{U} : d_2(u, \mathcal{U}_{\text{safe}}(x)) \leq r\}.$$

It is compact and convex, and for every nonempty G ,

$$\beta_{\text{geo}}(G) = \min\{r \geq 0 : \bigcap_{x \in G} B_x(r) \neq \emptyset\}.$$

The minimum exists because the supremum of the continuous distance functions is lower semicontinuous on compact $\mathcal{U}$.

Let b be the supremum of $\beta_{\text{geo}}(G)$ over subfibers of size at most $p+1$. Monotonicity under set inclusion gives $b \leq \beta_{\text{geo}}(F_z^R)$. If $r > b$, every at-most- $(p+1)$ subfamily of $\{B_x(r)\}_{x \in F_z^R}$ intersects. Helly's theorem makes every finite subfamily intersect; since every member is compact, the finite-intersection property makes the whole family intersect. Thus $\beta_{\text{geo}}(F_z^R) \leq r$. Letting $r \downarrow b$ proves the displayed equality. At $r = 0$, the same argument says that the full common-action set is nonempty exactly when every at-most- $(p+1)$ intersection is nonempty. Attainment and closedness make $\beta_{\text{geo}} = 0$ equivalent to a nonempty intersection. A finite fiber has only finitely many eligible subfibers, so its supremum is a maximum.

Sharpness follows with $B = \{u : \|u\|_2 \leq \sqrt{p}\}$,

$$A_i = B \cap \{u : u_i \geq 1\} \quad (1 \leq i \leq p), \quad A_{p+1} = B \cap \{u : \sum_i u_i \leq p-1\}.$$

The full intersection is empty. Omitting A_{p+1} leaves $(1, \dots, 1)$ feasible; omitting A_j leaves the vector with coordinate j zero and every other coordinate one. Hence every p -subfamily intersects and $p+1$ cannot be improved. $\square$

When $\mathcal{U}$ has affine dimension $q < p$, an affine isometry identifies its affine hull with $\mathbb{R}^q$; the same proof then replaces $p + 1$ by $q + 1$.

The assumptions delimit the result. In one dimension the three closed nonconvex sets $\{0, 1\}$, $\{1, 2\}$, $\{0, 2\}$ intersect pairwise but not jointly. Without compactness, an empty intersection need not have positive geometric slack: the ray $\{(t, 0) : t \geq 0\}$ and the closed convex epigraph $\{(t, s) : t > 0, s \geq 1/t\}$ are disjoint at distance zero. For an infinite noncompact family, $[n, +\infty)$, $n \geq 1$, has every finite intersection nonempty but empty total intersection. These examples rule out the corresponding stronger variants.

B.4 Finite-action slack and dual witness

Proof of Theorem 6. A Dirac mass on a deterministic minimizer gives $\beta_{\text{exp}} \leq \beta_{\text{det}}$. For any $p \in \Delta_k$, choose a_p with $p_{a_p} \geq 1/k$. Nonnegativity gives

$$\sup_x \sum_a p_a \ell(x, a) \geq p_{a_p} \sup_x \ell(x, a_p) \geq \frac{1}{k} \min_a \sup_x \ell(x, a).$$

Taking the infimum over p proves the sandwich. The action-wise robust objective is linear in p , so its minimum is attained at an action minimizing the robust loss. For any nonempty support, the maximum supported robust loss is at least the minimum across all actions, and a Dirac mass on a minimizer attains equality; hence $\beta_{\text{ar}} = \beta_{\text{ws}} = \beta_{\text{det}}$.

If the common safe-action set is empty and loss vanishes exactly on safe actions, every action has strictly positive loss at some compatible state. Finiteness of the action set makes $\beta_{\text{det}} > 0$, and the sandwich propagates positivity. Hard failure loss makes every action-wise supremum one. $\square$

When $F = \{x_1, \dots, x_n\}$ and $L_{ia} = \ell(x_i, a)$, finite linear-program duality gives

$$\beta_{\text{exp}}(z) = \max_{q \in \Delta_n} \min_a \sum_{i=1}^n q_i L_{ia}.$$

The primal minimizes t subject to $Lp \leq t\mathbf{1}$ and $p \in \Delta_k$; its dual maximizes r subject to $L^T q \geq r\mathbf{1}$ and $q \in \Delta_n$. The maximizing q is a finite adversarial mixture over compatible states, not a population certificate.

B.5 Smooth localized collision

Proof of Proposition 4. Let $\psi(t) = \exp(1 - (1 - t^2)^{-1})$ for $|t| < 1$ and zero otherwise, and define

$$R_\delta(x) = x - a\psi((x - a)/\delta) + a\psi((x + a)/\delta).$$

The two bump supports are disjoint and interior to $[-1, 1]$, while $R_\delta(a) = R_\delta(-a) = 0$. Thus a safe point of margin a collides with an unsafe point. With decoder and latent predictor equal to the identity, reconstruction error is supported on intervals of total length 4δ and has magnitude at most a ; the uniform density gives the bound $2a^2\delta$. Identity dynamics makes latent prediction exact, and decoded one-step prediction has the same error as reconstruction. Finally, a admits only action $+1$ and $-a$ only action -1 under the condition $ux \geq 0$. $\square$